Do Now · retrieval

Lesson 143 · The TCP/IP Stack and Encapsulation

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which device typically forwards traffic between different IP networks?

Q2

Which logic gate outputs 1 only when both inputs are 1?

Q3

What is the accumulator for?

Q4

Which is faster in the worst case on sorted arrays, linear or binary search?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Application data — HTTP sends a request body/application message.

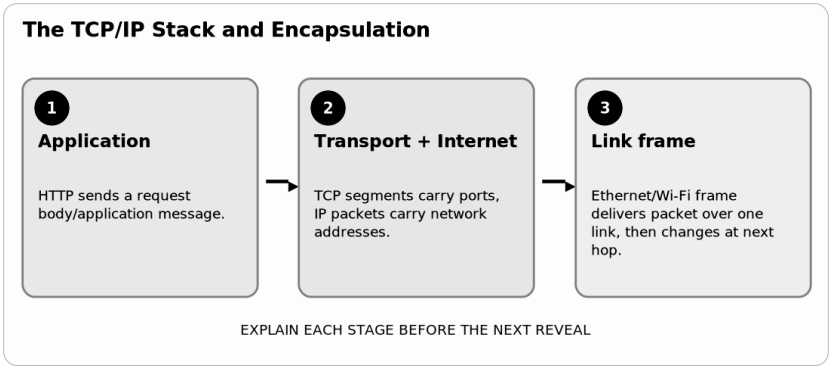
Core idea 2

Transport and IP — TCP segments carry ports, IP packets carry network addresses.

Core idea 3

Frame on local link — Ethernet/Wi-Fi frame delivers packet over one link, then changes at next hop.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

application layer — Protocols providing services to programs, such as HTTP or DNS.

transport layer — End-to-end service such as TCP or UDP.

internet layer — IP packet addressing and routing between networks.

link layer — Local frame delivery on physical or wireless link.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

A browser sends an HTTP request using TCP/IP over Ethernet. Describe encapsulation from top to bottom.

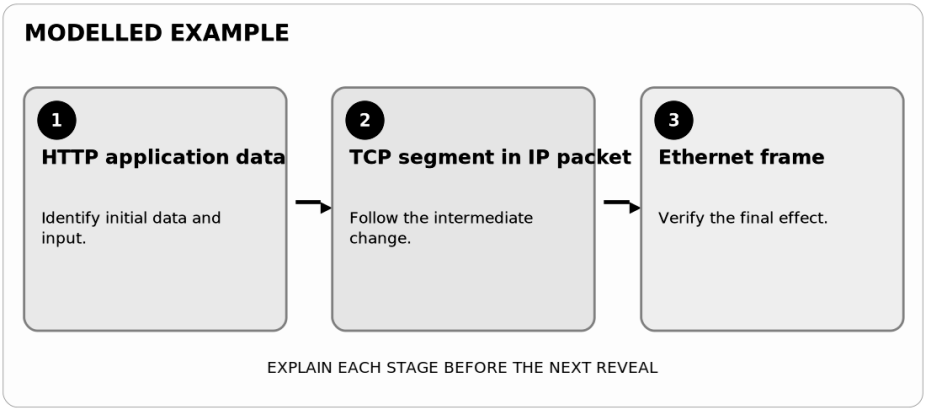
Follow the worked method

1. HTTP application data — Identify the relevant information and make a prediction.

2. TCP segment in IP packet — Explain the change using correct technical terms.

3. Ethernet frame — HTTP application bytes become TCP segment payload; the TCP segment goes into an IP packet; the packet is carried in an Ethernet frame. Receivers process headers in reverse to deliver data to the browser/server.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Incoming Ethernet frame
Identify the relevant information and make a prediction.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Name four layers of the TCP/IP stack and give one role for each. [8]

Question 2

Describe encapsulation of application data over TCP/IP and Ethernet. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain what a router changes when forwarding a packet to a different link technology. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Every router preserves the same link-layer frame all the way to the destination.

Use key vocabulary from memory

1. Explain application layer in your own words.

2. Explain transport layer in your own words.

3. Explain internet layer in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.